

Shaojie Hu

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RESEARCH INTERESTS

- **Phase equilibrium and transition in porous media:** Vapor sorption and capillary condensation, pore water freezing and melting, and pore salt deliquescence and efflorescence.
- **Interfacial thermodynamics and transport:** Solid–liquid interactions, film and confined water flows, and stable–isotope fractionation of pore water.
- **Poromechanics:** Unsaturated soil mechanics, soil water potential, and soil swelling and shrinkage.

EDUCATION

Ph.D., Geotechnical Engineering

Hunan University, Changsha, China

Supervisor: Prof. Chao Zhang

Sept. 2020 – Jul. 2026

B.S., Civil Engineering

Hunan University, Changsha, China

Sept. 2016 – Jun. 2020

PUBLICATIONS

Journal Articles (* Corresponding author)

- [1] **Hu, S.**, Zhang, C.*, Li, L., Or, D., Lu, N., & Chen, R. (2026). Fractionation due to soil–water interactions affects global isotopic evapotranspiration partitioning. *Advances in Water Resources*, 105406. [DOI link]
- [2] **Hu, S.**, Zhang, C.*, Li, L., & Lu, N. (2026). Determining soil water structure, density, and viscosity via infrared spectroscopy. *Journal of Engineering Mechanics*, 152(4), 04026002. [DOI link]
- [3] **Hu, S.**, Zhao, N., Zhang, C.*, Li, F., Chen, R., & Or, D.* (2025). Water nanofilms facilitate ice crystal growth across droplets. *Physical Review Letters*, 134(6), 064001. [DOI link]
- [4] **Hu, S.**, Zhang, C.*, Dong, Y., Gou, L., & Chen, R. (2025). Water vapor sorption isotherms of salt-affected soils. *Canadian Geotechnical Journal*, cgj-2024-0426. [DOI link]
- [5] **Hu, S.**, Zhang, C.*, & Lu, N. (2023). Quantifying coupling effects between soil matric potential and osmotic potential. *Water Resources Research*, 59(2), e2022WR033779. [DOI link]
- [6] **Hu, S.**, & Zhang, C.* (2023). A Sorption Isotherm Model for Soil Incorporating External and Internal Surface Adsorption, and Capillarity. *Canadian Geotechnical Journal*, cgj-2022-0386. [DOI link]
- [7] Zhang, C., Lin, X., **Hu, S.***, & Lu, N. (2025). Predicting the Soil Freezing Curve via a Modified Clapeyron Equation. *Journal of Geotechnical and Geoenvironmental Engineering*, 152(2), 04025195. [DOI link]
- [8] Zhang, C., Zhao, N., **Hu, S.***, & Lin, X. (2025). Salts retard ice crystal growth in supercooled droplets during recalescence. *Physical Review Fluids*, 10(6), 063605. [DOI link]

- [9] Gou, L., Lu, N., **Hu, S.**, Calderon, A. R. A., & Zhang, C.* (2025). Suction Stress of Soil Slurry. *Journal of Geotechnical and Geoenvironmental Engineering*, 151(2), 04024164. [DOI link]
- [10] Zhang, C.*, Zhao, N., **Hu, S.**, Chang, J., Miao, X., & Chen, R. (2025). Synchrotron-based WAXS–SAXS scattering of pore ice in clayey soil. *SCIENTIA SINICA Technologica*, 55(8), 1340–1350. (in Chinese). [DOI link]
- [11] Lin, X., Zhang, C.*, **Hu, S.**, & Chen, R. (2024). Heterogeneous ice nucleation of salt solution in porous media. *The Journal of Chemical Physics*, 160(9), 094501. [DOI link]
- [12] Zhang, C., Li, L., **Hu, S.***, Gou, L., & Chen, R. (2024). Physical origin of adsorption heat and its significance in the isotherm equation. *International Journal of Heat and Mass Transfer*, 220, 124914. [DOI link]
- [13] Zhao, N., **Hu, S.**, Zhang, C.*, Li, F., & Chen, R. (2023). Physical Origins of Freezing and Melting Temperature Depressions of Water in Millimeter-sized Pores. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 674, 131851. [DOI link]
- [14] Gou, L., Zhang, C.*, Lu, N., & **Hu, S.** (2023). A Soil Hydraulic Conductivity Equation Incorporating Adsorption and Capillarity. *Journal of Geotechnical and Geoenvironmental Engineering*, 149(8), 04023056. [DOI link]
- [15] Gou, L., Zhang, C.*, **Hu, S.**, Chen, R., & Dong, Y. (2023). Semi-analytical Solutions for Soil Consolidation Induced by Drying. *Acta Geotechnica*, 18(2), 739–755. [DOI link]
- [16] Zhang, C., **Hu, S.**, Qiu, Z., & Lu, N.* (2022). A Poroelasticity Theory for Soil Incorporating Adsorption and Capillarity. *Géotechnique*, 1–18. [DOI link]
- [17] Zhang, C., Gou, L., **Hu, S.***, & Lu, N. (2022). A Thermodynamic Formulation of Water Potential in Soil. *Water Resources Research*, 58(9). [DOI link]
- [18] Zhang, C.*, **Hu, S.**, & Lu, N. (2022). Unified Elastic Modulus Characteristic Curve Equation for Variably Saturated Soils. *Journal of Geotechnical and Geoenvironmental Engineering*, 148(1), 04021171. [DOI link]

Manuscripts under review and in preparation

- [19] **Hu, S.**, Zhang, C.*, Gou, L., & Chang, J. Activation Energy Model for Film Flow Incorporating Adsorption. *Under review*.
- [20] **Hu, S.**, et al. Thermodynamics of Equilibrium Isotope Fractionation of Water in Porous Media. *In preparation*.

PRESENTATIONS

Gordon Research Conference: Flow and Transport in Permeable Media

Les Diablerets, Switzerland

July 2026

Poster: Fractionation due to soil–water interactions affects global isotopic evapotranspiration partitioning

AGU25, New Orleans, LA, United States

December 2025

Poster: Fractionation Due to Soil–Water Interactions Is an Important Omission in Global Isotopic Evapotranspiration Partitioning

2025 Annual Conference of Chinese Society of Soil Science, Yantai, China

August 2025

Poster: Phase equilibrium and transitions in salt-affected soils

Ice at the Micro-scale 2025 , Ascona, Switzerland	July 2025
Poster: Salts retard ice crystal growth in supercooled droplets during recalescence	
AGU24 , Washington, D.C., United States	December 2024
Poster: Phase equilibrium and transitions in salt-affected soils	
InterPore2024 , Qingdao, China	May 2024
Oral: Adsorption-induced Effective Stress in Porous Media	
The 4th International Soil Modeling Consortium Conference , Tianjin, China	May 2024
Poster: Coupling Between Soil Matric Potential and Osmotic Potential	
InterPore2023 , Edinburgh, Scotland	May 2023
Poster: A Poroelasticity Theory for Soil Incorporating Adsorption and Capillarity	
The 9th Young Experts Forum on Geotechnical Engineering , Changsha, China	June 2021
Oral: Unified Elastic Modulus Function for Variably Saturated Soils	

ACADEMIC SERVICE

Editorial Service

Associate Editor, <i>Vadose Zone Journal</i>	2026 – present
Social Media Editor, <i>Vadose Zone Journal</i>	2025 – 2026

Journal Reviewer

Communications Earth & Environment
Water Resources Research
Vadose Zone Journal
Geoscientific Model Development
Bulletin of Engineering Geology and the Environment
Construction and Building Materials
Geotechnical Research

AWARDS & HONORS

First Prize of AGU Student Award	2024
Unsaturated Zone Technical Committee, American Geophysical Union	
National Scholarship for Graduate Students	2024
Ministry of Education, China	
Engineering Innovation Design Scholarship	2022
Hunan University	
Outstanding Innovative Undergraduate Thesis	2020
Hunan University	
First-Class Scholarship for Undergraduate Students	2018
Hunan University	